

HARCOURT BUTLER TECHNICAL UNIVERSITY (HBTU), KANPUR

B.Tech 1st Year • Engineering Mechanics (Unit 5)

Introduction to Strength of Materials (Mechanics of Deformable Bodies)

Complete Academic Notes, Theoretical Formulations, Vector Schematics, Multi-Axial Stress & Principal Strains

100% Complete Syllabus

Zero Topic Left

Clean Notation & Formulae

HBTU Examination Aligned

PRESCRIBED SYLLABUS TOPICS COVERED IN THIS DOCUMENT

- **Introduction to Strength of Materials:** Rigid vs. Deformable bodies, Assumptions
- **Stresses & Strains:** Normal, Shear, Volumetric, Bearing stresses and complementary shear
- **Stress-Strain Curves:** Mild steel (ductile) with all 6 salient points & Cast iron (brittle)
- **Elastic Constants:** Young's modulus (E), Shear modulus (G), Bulk modulus (K), Poisson's ratio (μ)
- **Interrelationships of Elastic Constants:** Complete derivations and bounds of Poisson's ratio
- **1D Loading of Varying Cross-Sections:** Stepped bars, Uniformly tapering circular and rectangular bars
- **Bars of Uniform Strength & Self-Weight:** Prismatic elongation and exponential profile
- **Strain Energy:** Resilience, Proof resilience, Modulus of resilience, Gradual vs. Sudden/Impact loading
- **2D State of Plane Stress:** Stresses on oblique planes, Transformation equations & Mohr's Circle
- **Principal Stresses, Planes & Strains:** Maximum shear stress, Generalized Hooke's Law & Volumetric strain

1. Introduction to Strength of Materials

While **Engineering Mechanics (Statics and Dynamics)** investigates the external equilibrium and motion of strictly *rigid bodies* (idealized bodies whose internal particles experience zero relative displacement under external loads), practical engineering structures, bridges, and machine components deform under applied forces.

Definition of Strength of Materials (Mechanics of Deformable Bodies): It is the branch of applied engineering mechanics that investigates the internal resisting forces (stresses), geometric deformations (strains), and structural stability of deformable solid bodies subjected to external mechanical and thermal loads. Its primary objective is to evaluate load-carrying capacity, prevent catastrophic yielding or brittle fracture, and ensure safe, economical structural design.

1.1 Fundamental Assumptions in Strength of Materials

1. **Continuity:** The material is completely continuous throughout its volume with no microscopic voids, cracks, or cavities.
2. **Homogeneity:** The physical and mechanical properties are identical at all points within the body.
3. **Isotropy:** The elastic properties of the material are independent of direction (identical in all spatial orientations).
4. **Elasticity and Hooke's Law:** Deformations are completely recoverable upon removal of external loads, and stress is directly proportional to strain up to the proportional limit.
5. **Saint-Venant's Principle:** The localized stress distributions produced by statically equivalent load systems differ only in the immediate vicinity of the loading points; at distances comparable to the greatest cross-sectional dimension, internal stresses depend solely on the resultant force and bending moment.
6. **Small Deformation Theory:** Strains and displacements are sufficiently small such that the initial un-deformed geometry can be utilized in equilibrium equations.

2. Normal and Shear Stresses and Strains

2.1 Concept of Stress (Internal Resisting Force)

When an external load acts on a deformable body, internal intermolecular resisting forces are generated across every internal cross-section to maintain mechanical equilibrium. Stress is defined as the internal resisting force developed per unit area of the cross-section.

$$\sigma = \lim_{(\Delta A \rightarrow 0)} (\Delta P / \Delta A) = P / A \quad [SI \text{ Unit: } N/m^2 \text{ or Pascal (Pa)}]$$

In standard mechanical engineering design: 1 MPa = 10^6 N/m² = 1 N/mm² and 1 GPa = 10^9 N/m² = 10^3 N/mm².

2.2 Classification of Stresses

- **Normal Stress (σ):** Acts perpendicular (normal) to the plane of the cross-section.
 - *Tensile Stress (σ_t):* Pulls away from the cross-section, causing axial elongation ($\Delta L > 0$).
 - *Compressive Stress (σ_c):* Pushes into the cross-section, causing axial shortening ($\Delta L < 0$).
- **Shear Stress (τ):** Acts tangential or parallel to the plane of the cross-section, tending to slide one internal layer over another.

$$\tau = P_{\text{tangential}} / A = V / A$$

Examples: Riveted and bolted joints undergoing single or double shear, punch press operations, torque transmission in drive shafts.

- **Bearing Stress (σ_b):** Localized compressive contact pressure developed at the physical interface between two mating surface bodies (e.g., pin resting in a clevis hole, foundation footing).

$$\sigma_b = P / (d \cdot t) \quad (\text{Projected area } A_p = d \cdot t)$$

2.3 Complementary Shear Stress Theorem

Theorem: A shear stress acting on a given plane is always accompanied by an equal and opposite shear stress acting on an adjacent mutually perpendicular plane to satisfy rotational moment equilibrium of the differential element.

Consider an infinitesimal cubical element of dimensions $dx \times dy \times dz$. If a tangential shear stress τ_{xy} acts on the top and bottom faces, the resulting clockwise couple is $(\tau_{xy} \cdot dx \cdot dz) \cdot dy$. To balance this couple, counter-clockwise shear stresses τ_{yx} must act on the vertical faces such that $(\tau_{yx} \cdot dy \cdot dz) \cdot dx = (\tau_{xy} \cdot dx \cdot dz) \cdot dy$. Hence:

$$\tau_{xy} = \tau_{yx}$$

2.4 Concept of Strain

Strain is a dimensionless measure of the geometric deformation experienced by a body under applied stress.

- **Longitudinal / Normal Strain (ϵ):** Change in length per unit original length:

$$\epsilon = \Delta L / L_0 \quad (\text{Dimensionless, often expressed as } \mu\text{m/m or } \%)$$

- **Shear Strain (γ):** The angular distortion or change in the right angle (90°) between two mutually perpendicular reference lines on an element:

$$\gamma = \tan \phi \approx \phi \quad (\text{measured in radians})$$

- **Volumetric Strain (ϵ_v):** Fractional change in the total volume of a body under multi-axial stress:

$$\epsilon_v = \Delta V / V_0 = \epsilon_x + \epsilon_y + \epsilon_z$$

3. Stress-Strain Diagrams for Ductile and Brittle Materials

3.1 Stress-Strain Curve for Mild Steel (Ductile Material)

When a standard circular mild steel specimen is tested under quasi-static uniaxial tension on a Universal Testing Machine (UTM) complying with ASTM/IS standards, a characteristic stress-strain response is recorded.

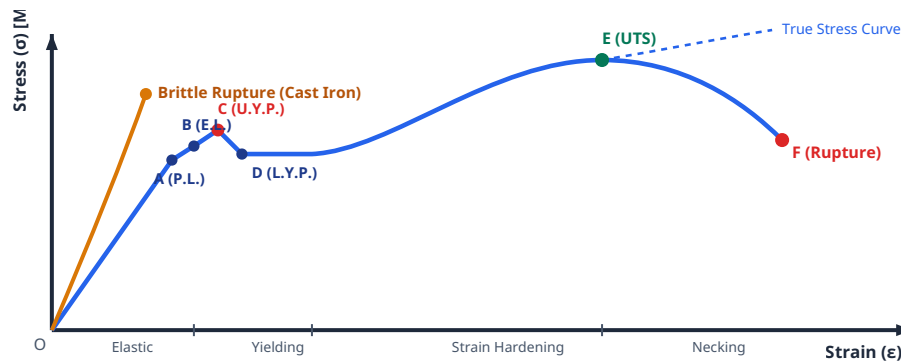


Figure 3.1: Engineering Stress-Strain Diagrams for Ductile Mild Steel vs. Brittle Cast Iron

3.2 Detailed Analysis of Salient Points (Mild Steel)

Salient Point	Physical Definition	Microstructural Mechanism & Engineering Significance
Point A: Proportional Limit	Highest stress up to which stress is directly proportional to strain ($\sigma \propto \epsilon$).	Hooke's Law is rigorously valid. The linear slope represents Young's Modulus of Elasticity (E) . Atomic lattice bonds stretch elastically.
Point B: Elastic Limit	Maximum stress that the material can sustain without undergoing permanent plastic deformation.	Upon complete unloading from point B, the specimen returns along path BO to its original zero-strain geometry. Beyond B, permanent plastic set initiates.
Point C: Upper Yield Point	Point where plastic deformation initiates suddenly with an unstable drop in required load.	Dislocations break free from interstitial carbon/nitrogen Cottrell atmospheres, resulting in sudden slip. Highly sensitive to testing rate and specimen alignment.
Point D: Lower Yield Point	Constant stress plateau where plastic strain increases significantly without any increase in load.	Lüders bands propagate across the specimen gauge length. In structural engineering design, the Yield Strength ($\sigma_y \approx 250$ MPa) is taken as the primary failure criterion.
Point E: Ultimate Tensile Strength (UTS)	Maximum engineering stress the specimen can withstand over its original cross-sectional area A_0 .	Substantial plastic deformation leads to dislocation entanglement and strain hardening (work hardening) . Peak load carrying capacity ($\sigma_u \approx 410\text{--}450$ MPa).
Point F: Fracture / Rupture Point	Stress at which physical severance and complete separation of the specimen occurs.	Beyond UTS, deformation localizes into severe necking ($A_{\text{actual}} \ll A_0$). The specimen fails via a classic Cup-and-Cone fracture at 45° shear plane.

3.3 Stress-Strain Behavior of Brittle Materials (Cast Iron, Concrete, Glass)

Characteristics of Brittle Failure:

- Brittle materials exhibit very small or imperceptible plastic deformation before sudden catastrophic fracture ($\epsilon_f < 5\%$).
- There is **no well-defined yield point**, no yield plateau, and no necking phenomenon.
- Tensile strength is dramatically lower than compressive strength (Grey cast iron is approximately 3 to 4 times stronger in compression than in tension).
- Failure occurs on a plane perpendicular to the tensile load axis (90° flat cleavage plane governed by maximum principal tensile stress).
- **0.2% Offset Proof Stress:** Since brittle materials (and non-ferrous metals like Aluminum) lack a distinct yield point, a line is drawn parallel to the elastic slope from a strain of $\epsilon = 0.002$ (0.2%). The intersection point defines the **Proof Yield Stress ($\sigma_{0.2}$)**.

4. Elastic Constants and Their Interrelationships

For an isotropic, homogeneous, linearly elastic material, mechanical deformation behavior is characterized by four fundamental **Elastic Constants**.

4.1 Definitions of the Four Elastic Constants

1. **Young's Modulus of Elasticity (E):** Ratio of longitudinal normal stress (σ) to longitudinal normal strain (ϵ) within proportional limit:

$$E = \sigma / \epsilon \quad [SI \text{ Unit: } N/m^2, Pa, \text{ or } GPa]$$

Represents inherent axial stiffness (For structural steel, $E \approx 200\text{--}210$ GPa).

2. **Shear Modulus / Modulus of Rigidity (G or C):** Ratio of shear stress (τ) to shear strain (γ) within elastic limit:

$$G = \tau / \gamma \quad [SI \text{ Unit: } N/m^2, Pa, \text{ or } GPa]$$

Represents torsional and shear stiffness (For steel, $G \approx 78\text{--}84$ GPa).

3. **Bulk Modulus (K):** Ratio of uniform volumetric hydrostatic stress ($\sigma_h = P$) to volumetric strain (ϵ_v):

$$K = \sigma_h / \epsilon_v = -P / (\Delta V / V_0) \quad [SI \text{ Unit: } N/m^2, Pa, \text{ or } GPa]$$

Measures incompressibility or resistance to volume change under uniform hydrostatic pressure.

4. **Poisson's Ratio (μ or ν):** When an axial load causes longitudinal extension, it produces simultaneous lateral contraction. Poisson's ratio is the negative ratio of lateral (transverse) strain to longitudinal (axial) strain:

$$\mu = - (\text{Lateral Strain} / \text{Longitudinal Strain}) = - (\epsilon_{lat} / \epsilon_{long})$$

- For steel: $\mu \approx 0.28\text{--}0.33$.
- For concrete: $\mu \approx 0.1\text{--}0.2$; Cork: $\mu \approx 0$; Incompressible rubber: $\mu = 0.5$.
- Theoretical bounds for isotropic materials: $-1.0 \leq \mu \leq 0.5$ (practically $0 \leq \mu \leq 0.5$).

4.2 Governing Interrelationships of Elastic Constants

The fundamental mathematical equations connecting E, G, K, and μ derived from multi-axial Hooke's Law:

$$E = 2G (1 + \mu)$$

$$E = 3K (1 - 2\mu)$$

$$E = (9KG) / (3K + G) \quad \text{implies} \quad 9 / E = (3 / G) + (1 / K)$$

$$\mu = (3K - 2G) / (6K + 2G)$$

Number of Independent Elastic Constants:

- For **Isotropic & Homogeneous** material: Out of 4 constants (E, G, K, μ), exactly **2 are independent**.
- For **Orthotropic** material (different properties along 3 mutually perpendicular planes, e.g., wood): **9 independent constants**.
- For **Anisotropic / Triclinic** material: **21 independent constants**.

5. One-Dimensional Loading of Members of Varying Cross-Sections

5.1 Prismatic Bar Subjected to Axial Load P

For a bar of uniform cross-sectional area A, length L, and modulus E under axial force P:

$$\sigma = P/A, \quad \varepsilon = \Delta L/L = \sigma/E \implies \Delta L = (PL)/(AE)$$

5.2 Stepped Bar (Compound Axial Member)

A member consisting of n distinct cylindrical or prismatic segments with areas A_i , lengths L_i , and Young's moduli E_i carrying axial force P_i :

$$\Delta L_{\text{total}} = \sum_{(i=1 \text{ to } n)} [(P_i L_i) / (A_i E_i)] = [(P_1 L_1) / (A_1 E_1)] + [(P_2 L_2) / (A_2 E_2)] + \dots + [(P_n L_n) / (A_n E_n)]$$

5.3 Uniformly Tapering Circular Bar

Consider a circular bar of length L tapering linearly from diameter d_1 at one end to d_2 at the other under axial pull P . At distance x from the smaller end:

$$d_x = d_1 + [(d_2 - d_1) / L] x \implies A_x = (\pi/4) d_x^2$$

Elongation of an elemental disc of thickness dx :

$$d(\Delta L) = (P dx) / (A_x E) = (4P dx) / (\pi [d_1 + kx]^2 E), \quad \text{where } k = (d_2 - d_1) / L$$

Integrating over length from $x = 0$ to $x = L$:

$$\Delta L = \int_0^L [(4P dx) / (\pi E [d_1 + kx]^2)] = (4PL) / (\pi E d_1 d_2)$$

Key Result: The elongation of a circular tapering bar is identical to that of an equivalent prismatic bar having effective diameter $d_{\text{eff}} = \sqrt{(d_1 d_2)}$ (geometric mean of end diameters).

5.4 Uniformly Tapering Rectangular Bar

Consider a plate of constant thickness t , length L , whose width tapers linearly from b_1 to b_2 under axial force P :

$$b_x = b_1 + [(b_2 - b_1) / L] x \implies A_x = b_x \cdot t$$

Integrating the elemental elongation:

$$\Delta L = \int_0^L [(P dx) / (t b_x E)] = [(PL) / (t E [b_2 - b_1])] \cdot \ln(b_2 / b_1)$$

5.5 Elongation of a Bar Due to Self-Weight

- **Prismatic Bar Suspended Vertically:** Density ρ , specific weight $\gamma = \rho g$, length L , cross-section A .

At distance x from the free bottom end, the hanging weight is $W_x = \gamma A x$.

$$\Delta L = \int_0^L [(\gamma A x dx) / (A E)] = (\gamma L^2) / (2E) = (W L) / (2 A E)$$

Crucial Insight: Elongation due to self-weight is exactly **half (50%)** of the elongation produced by an external concentrated load equal to the total weight (W) applied at the bottom end.

- **Bar of Uniform Strength:** A vertical hanging member whose cross-sectional area A_x decreases from top support to bottom free end such that normal stress $\sigma = \text{constant}$ at all sections:

$$A_x = A_0 \exp[(\gamma x) / \sigma] = A_0 e^{[(\rho g x) / \sigma]}$$

6. Strain Energy & Impact Loading

6.1 Concept of Strain Energy (Internal Work Stored)

When an external force deforms an elastic member, work is performed. In the absence of kinetic energy changes or thermal dissipation, this work is completely converted and stored within the deformed body as internal potential energy, termed **Strain Energy (U)**.

$$U = (1/2) P \Delta L = (1/2) P [(PL) / (AE)] = (P^2 L) / (2AE) = [\sigma^2 / (2E)] \times (A \cdot L)$$

Strain energy stored per unit volume is known as **Strain Energy Density (u)**:

$$u = U / V = \sigma^2 / (2E) = (1/2) \sigma \epsilon \quad [\text{Area under the elastic } \sigma\text{--}\epsilon \text{ curve}]$$

6.2 Key Energy Terminology

Energy Metric	Mathematical Formulation	Physical Definition & Application
Resilience	$U = [\sigma^2 / (2E)] \cdot V$	Capacity of a body to store strain energy within the elastic limit and release it completely upon unloading.
Proof Resilience	$U_p = [\sigma_e^2 / (2E)] \cdot V$	The maximum strain energy stored up to the elastic limit (σ_e). Represents peak elastic shock absorption.
Modulus of Resilience	$u_r = \sigma_e^2 / (2E)$	Proof resilience per unit volume. Governs material selection for mechanical springs and vehicle suspension dampers.
Modulus of Toughness	$u_t = \int_0^{\epsilon_f} \sigma \, d\epsilon$	Total energy absorbed per unit volume up to catastrophic fracture. Equal to total area under complete σ – ϵ curve.

6.3 Types of Axial Loading: Gradual, Sudden, and Impact

1. **Gradually Applied Load:** Load increases infinitesimally from 0 to P.

$$\sigma_{grad} = P / A, \quad \Delta L_{grad} = (PL) / (AE)$$

2. **Suddenly Applied Load:** Full load P applied instantaneously with zero dropping height ($h = 0$).

Equating external work done $W_{ext} = P \cdot \Delta L$ to strain energy $U = [\sigma^2 / (2E)] AL$:

$$P \cdot \Delta L = [\sigma^2 / (2E)] AL \implies P [(\sigma L) / E] = [\sigma^2 / (2E)] AL \implies \sigma_{sudden} = (2P) / A = 2 \sigma_{grad}$$

Crucial Exam Point: A suddenly applied load produces exactly **twice the stress** and twice the instantaneous elongation of an identical gradually applied load.

3. **Impact / Falling Load:** Load P dropped from height h onto a collar fixed to the end of a vertical bar.

$$\text{Work Done} = P(h + \Delta L) = [\sigma_i^2 / (2E)] AL$$

Solving the resulting quadratic equation in impact stress σ_i :

$$\sigma_i = \sigma_{st} [1 + \sqrt{1 + [(2h) / (\Delta L_{st})]}]$$

where $\sigma_{st} = P / A$ is the static stress, and $\Delta L_{st} = (PL) / (AE)$ is the static deflection.

The **Impact Factor (IF)** is defined as:

$$\text{IF} = \sigma_i / \sigma_{st} = 1 + \sqrt{1 + [(2h) / (\Delta L_{st})]}$$

When $h = 0$, $IF = 1 + \sqrt{1 + 0} = 2$ (recovering the suddenly applied case).

7. Two-Dimensional (2D) State of Plane Stress

7.1 General Biaxial Stress State on an Infinitesimal Element

Consider an infinitesimal element subjected to normal stresses σ_x and σ_y on mutually perpendicular planes and shear stresses $\tau_{xy} = \tau_{yx}$.

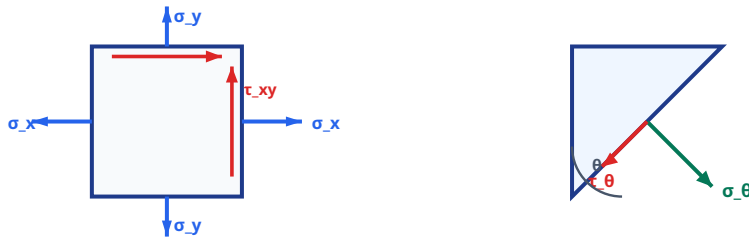


Figure 7.1: General 2D State of Plane Stress and Transformation onto an Oblique Plane at Angle θ

7.2 Stresses on an Oblique Plane (Analytical Transformation)

Consider an inclined cut plane whose normal makes a counter-clockwise angle θ with the x-axis. From static force equilibrium of the triangular wedge:

$$\sigma_{\theta} = [(\sigma_x + \sigma_y) / 2] + [(\sigma_x - \sigma_y) / 2] \cos 2\theta + \tau_{xy} \sin 2\theta$$

$$\tau_{\theta} = -[(\sigma_x - \sigma_y) / 2] \sin 2\theta + \tau_{xy} \cos 2\theta$$

Stress Invariant I: The sum of normal stresses on any two mutually perpendicular planes (θ and $\theta + 90^\circ$) is constant:

$$\sigma_{\theta} + \sigma_{\theta + 90^\circ} = \sigma_x + \sigma_y = \text{ext}\{\text{Invariant}\}$$

8. Principal Stresses, Principal Planes, and Mohr's Circle

8.1 Principal Planes & Principal Stresses

Definition: Principal Planes are planes of zero shear stress ($\tau_{\theta} = 0$). The normal stresses acting on these planes are stationary (extreme values) and are termed **Principal Stresses** (σ_1, σ_2).

Orientation of Principal Planes

Setting $\tau_{\theta} = 0$ in the transformation equation yields:

$$\tan 2\theta_p = (2\tau_{xy}) / (\sigma_x - \sigma_y)$$

This equation yields two roots for $2\theta_p$ separated by 180° , meaning the two principal planes θ_{p1} and θ_{p2} are **mutually perpendicular (90° apart)**.

Analytical Formula for Major and Minor Principal Stresses

$$\sigma_{1, 2} = [(\sigma_x + \sigma_y) / 2] \pm \sqrt{[(\sigma_x - \sigma_y) / 2]^2 + \tau_{xy}^2}$$

where σ_1 is the Major Principal Stress (maximum normal stress), and σ_2 is the Minor Principal Stress (minimum normal stress).

8.2 Maximum In-Plane Shear Stress ($\tau_{\max}$)

To find the plane of maximum shear stress, set $(d\tau_\theta / d\theta) = 0$:

$$\tan 2\theta_s = -(\sigma_x - \sigma_y) / (2\tau_{xy}) = -1 / (\tan 2\theta_p) \implies 2\theta_s = 2\theta_p \pm 90^\circ \implies \theta_s = \theta_p \pm 45^\circ$$

Vital Engineering Rules:

- The planes of maximum shear stress are inclined at exactly **45° to the principal planes**.
- On the plane of maximum shear stress, the normal stress is *not zero*; it equals the average normal stress:

$$\sigma_{\text{avg}} = (\sigma_x + \sigma_y) / 2 = (\sigma_1 + \sigma_2) / 2$$

- The magnitude of maximum in-plane shear stress is:

$$\tau_{\max} = (\sigma_1 - \sigma_2) / 2 = \sqrt{[(\sigma_x - \sigma_y) / 2]^2 + \tau_{xy}^2}$$

8.3 Mohr's Circle for Plane Stress (Graphical Construction)

Mohr's circle is a geometric representation of 2D stress transformation equations:

$$[\sigma_\theta - (\sigma_x + \sigma_y) / 2]^2 + \tau_\theta^2 = [(\sigma_x - \sigma_y) / 2]^2 + \tau_{xy}^2 = R^2$$

This is the standard equation of a circle in the (σ, τ) plane: $(x - h)^2 + y^2 = R^2$.

- **Center of Mohr's Circle (C):**

$$C = ((\sigma_x + \sigma_y) / 2, 0)$$

- **Radius of Mohr's Circle (R):**

$$R = \sqrt{[(\sigma_x - \sigma_y) / 2]^2 + \tau_{xy}^2} = \tau_{\max} = (\sigma_1 - \sigma_2) / 2$$

- **Sign Convention:** Tensile normal stress is positive (plotted to the right of origin). Shear stress producing a clockwise couple on the element face is plotted above the σ -axis (+ve), and counter-clockwise below (-ve). An angle of θ on the physical element corresponds to 2θ in the same direction on Mohr's circle.

8.4 2D Principal Strains & Generalized Hooke's Law

For a biaxial plane stress state ($\sigma_z = 0$), strains are determined using superposition of Poisson's effects:

$$\epsilon_x = (\sigma_x / E) - \mu (\sigma_y / E), \quad \epsilon_y = (\sigma_y / E) - \mu (\sigma_x / E), \quad \epsilon_z = -\mu [(\sigma_x + \sigma_y) / E]$$

Principal Strains (ϵ_1, ϵ_2):

$$\epsilon_{1,2} = [(\epsilon_x + \epsilon_y) / 2] \pm \sqrt{[(\epsilon_x - \epsilon_y) / 2]^2 + (\gamma_{xy} / 2)^2}$$

Maximum Shear Strain:

$$(\gamma_{\max} / 2) = (\epsilon_1 - \epsilon_2) / 2 \implies \gamma_{\max} = \epsilon_1 - \epsilon_2$$

9. Rapid Revision Formula Sheet & Examination Summary

Subject Topic	Fundamental Principles	Key Formulas & Governing Equations
Normal & Shear Stresses	Stress = Internal resistance / Area. Complementary shear: $\tau_{xy} = \tau_{yx}$.	$\sigma = P / A$, $\tau = V / A$, $\sigma_b = P / (d \cdot t)$
Elastic Constants	E: Axial stiffness; G: Shear stiffness; K: Volumetric stiffness; μ : Lateral contraction.	$E = 2G(1 + \mu) = 3K(1 - 2\mu)$ $E = (9KG) / (3K + G)$, $\mu = (3K - 2G) / (6K + 2G)$
Axial Deformations (ΔL)	Stepped bars, tapering bars, and self-weight effects.	Stepped: $\Delta L = \sum [(P_i L_i) / (A_i E_i)]$ Circular Taper: $\Delta L = (4PL) / (\pi E d_1 d_2)$ Self-Weight: $\Delta L = (\gamma L^2) / (2E) = (WL) / (2AE)$
Strain Energy & Loading	Work stored during elastic deformation. Gradual vs. Sudden vs. Impact.	$U = [\sigma^2 / (2E)] V$, $u_r = \sigma_e^2 / (2E)$ $\sigma_{\text{sudden}} = 2 \sigma_{\text{grad}}$ $IF = 1 + \sqrt{1 + [(2h) / (\Delta L_{st})]}$
Plane Stress Transformation	Normal and shear stresses on oblique plane at angle θ .	$\sigma_\theta = [(\sigma_x + \sigma_y) / 2] + [(\sigma_x - \sigma_y) / 2] \cos 2\theta + \tau_{xy} \sin 2\theta$ $\tau_\theta = - [(\sigma_x - \sigma_y) / 2] \sin 2\theta + \tau_{xy} \cos 2\theta$
Principal Stresses & Mohr's Circle	Planes of zero shear stress ($\tau = 0$). Planes of $\tau_{\max}$ at 45° to principal planes.	$\sigma_{1, 2} = [(\sigma_x + \sigma_y) / 2] \pm \sqrt{ [(\sigma_x - \sigma_y) / 2]^2 + \tau_{xy}^2 }$ $\tau_{\max} = (\sigma_1 - \sigma_2) / 2 = R$ Center: $([\sigma_x + \sigma_y] / 2, 0)$

HBTU Semester Examination Strategy for Unit 5:

- For numericals on varying cross-sections, always identify whether the load is axial concentrated or distributed self-weight before setting up the differential integral $d(\Delta) = (P_x dx) / (A_x E_x)$.
- In plane stress numericals, clearly write the given stress state with signs ($\sigma_x, \sigma_y, \tau_{xy}$), calculate the center C and radius R, and state the orientation of principal planes ($2\theta_p$) with proper counter-clockwise/clockwise sense.